

Monthly Intelligence Report

Edition 001 — Session 1 — Greenland

First scored edition of the Greenland fleet — 37 substations, 74 plants and 125 power-line segments across all 5 Kommuner, scored 2026-04-16 with Arctic/Greenland-calibrated inputs. Fleet median $R_{\text{median}} = 0.779$ · 29 of 37 substations in the Critical band. Arctic R6a restoration ($\alpha \in [0.55, 0.90]$) and GEUS stable-craton R6b ($\alpha \in [0.05, 0.15]$) live. SSI v4.0.2 · April 2026.

OFFICIAL · EDITION 001 · SESSION 1 SCORED 2026-04-16

data_status = OFFICIAL · fields_scored = 95/95 · MC = 1,000 × 37 = 37,000 draws

Every component, modifier and Monte Carlo interval is populated. Geometry: 37 substations across Kujalleq (20), Qeqertalik (12), Qeqqata (3), Sermersooq (1) and Avannaata (1 — Ilulissat transformer, OSM node/11958572184). Scoring is calibrated against Stat.gl 2024/25 socio-economic data, Nukissiorfiit 2024 Annual Report (CAIDI ≈ 8h, RE share ≈ 70%), Asiaq/DMI climate trajectories and GEUS stable-craton seismic. §8 Small-N guardrail is active (N=37 · R3 tier-balance 2–50% · no Gaussian copula — see Methodology §8.2). The Session 1 gate was brought forward from 2026-07 to 2026-04; standard monthly cadence begins 2026-05.

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SECTION A

Fleet Summary — Session 1

SUBSTATIONS SCORED

37

36 OSM + 1 Avannaata (Ilulissat)

FLEET MEDIAN R

0.779

Critical band (0.75–1.00)

CRITICAL-BAND SHARE

78%

29 of 37 substations

ACTIVE KOMMUNER

5

of 5 · Avannaata now scored

The SSI Index's Greenland Edition 001 is the first scored pass over a publicly-verifiable fleet. Session 1 comprises **37 substations, 74 plants / generators and 125 power-line segments** — every asset geolocated from OpenStreetMap (extract 2026-04-16), with Pituffik Space Base (U.S. Navy / Space Force / Air Force operator, lat 75.5°–77.5°N, lon -71°...

-66°) excluded by policy (see methodology §7). No synthetic nodes, no inferred cross-settlement edges: Greenland's grid is a set of isolated within-island networks and the topology reflects that exactly. The Avannaata coverage gap previously flagged at bootstrap was closed by direct OSM Overpass retrieval of node/11958572184 (Ilulissat transformer, 69.2205°N, -51.1012°E, 10.5 kV).

Regional distribution of the scored fleet and Kommune-level R_median: **Avannaata** 1 (R = 0.868), **Qeqertalik** 12 (R = 0.797), **Kujalleq** 20 (R = 0.774), **Qeqqata** 3 (R = 0.677), **Sermersooq** 1 (R = 0.655). The north-to-south gradient reflects Arctic restoration penalty (R6a $\alpha \in [0.55, 0.90]$), permafrost-degradation infrastructure wear, CAIDI $\approx 8h$ (vs Nordic $\sim 2h$), and socio-economic vulnerability from Stat.gl 2024 (elderly %, unemployment, GDP/capita). Sermersooq scores best because Nuuk is anchored by 45 MW Buksefjord hydro and the metro load concentration lowers R3 saturation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 25+ verified public sources (Nukissiorfiit, Departementet for Boliger og Infrastruktur, Grønlands Statistik, Asiaq, GEUS, OpenStreetMap, CMIP6, Open-Meteo). The §8 Small-N guardrail runs the Monte Carlo at 1,000 iterations per substation (37,000 total draws) with components perturbed independently — the full 95-dimensional Gaussian copula is held off at N=37 because empirical correlation matrices inferred from 37 observations are numerically unstable. The copula is scheduled to activate once the Nukissiorfiit asset register supplements OSM and N crosses 500.

SECTION B

Asset in Focus

Each edition spotlights one substation. For Edition 001 the spotlight rotates deterministically on a month-seed through the scored fleet — the card below is populated live from ssi-data.json and shows the Monte Carlo R_median, P5/P95 confidence interval, classification band, fleet percentile, and the full C/V/I/E/S/T component stack plus R3/R4/R6a/R6b/R7 modifier chain.

Substation in Focus

Qaqortoq transformerstation

Kujalleq · 63 kV

0.587

High

±0.095

P8

R_MEDIAN

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE

COMPONENTS (C · V · I · E · S · T)		MODIFIERS (R3 · R4 · R6A · R6B · R7)	GEOMETRY & MC
C: 0.565		R3: —	id: GL_0023
V: 0.623		R4: —	operator: Nukissiorfiit
I: 0.577		R6a: —	coord: 60.7144, -46.0410
E: 0.638		R6b: —	degree: 3 · PGA: 0.02
S: 0.624		R7: —	MC: 1,000 iter · R∈[0.487, 0.677]
T: 0.228			

This edition's spotlight — automatically selected for **April 2026** — is **Qaqortoq transformerstation** in Kujalleq, operated by Nukissiorfiit. $R_median = 0.587$ places it in the **High** band (fleet P8). The dominant component is **E** (0.638), and the heaviest modifier is **R6_restoration** ($\times 1.100$). Monte Carlo ran 1,000 iterations at the §8 Small-N setting, yielding a CI width of 0.190.

SECTION C — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component cross these silos — identifying substations where low digital resilience intersects with high socio-economic consequence. This is precisely the anticipatory intelligence that Nukissiorfiit and Departementet for Boliger og Infrastruktur strategic planning will require.

C.1 — Cyber-Physical Risk Matrix

R7 digital-readiness modifier vs overall R_median . At Edition 001 R7 is calibrated against the ENISA NIS2 small-utility cyber baseline ($V1 \approx 0.65$) because Greenland Connect's 2024 broadband coverage and Nukissiorfiit's OT segmentation are both weak signals on a 37-node fleet. R7 is uniformly in the [1.04, 1.08] band across substations — a flat digital penalty. The blind-spot quadrant (high R_median · high R7 penalty) therefore tracks R_median directly: every substation above 0.85 R_median doubles as a cyber-exposure candidate.

Top blind-spot candidates: the 29 Critical-band substations — with Avannaata (Ilulissat transformer) $R=0.868$ as the single highest-risk node. Full matrix will expand once Nukissiorfiit smart-meter penetration and OT segmentation indicators ingest at Edition 002.

C.2 — Economic Impact by Industrial Sector

R3 consequence segmentation, Edition 001: **Hydro & heavy industry** — Buksefjord-fed Sermersooq nodes (1), $R3 \approx 1.18$. **Commercial & Nuuk metro** — Sermersooq/Qeqqata settlement clusters (4), $R3 \approx 1.15$. **Settlement** — Kujalleq + Qeqertalik micro-grids (31), $R3 \approx 1.25$ (elderly % 13–16, unemployment 11–13%, GDP/cap DKK 205–235k). **Rural / remote** — Avannaata Ilulissat isolated transformer (1), $R3 \approx 1.28$.

VoLL range (indicative, 2024 DKK/MWh): hydro & heavy 22–40k · commercial 11–16k · settlement 6–9k · rural 3–5k. Greenland's R3 band overall is compressed high because every kommune is below the pan-OECD median for socio-economic resilience.

C.3 — Kommuner Digital Readiness Ranking

Average R7 by Kommune (lower R7 = better digital readiness $\Rightarrow$ smaller penalty multiplier): **Sermersooq** 1.04 (Nuuk fibre + smart-meter pilots) · **Qeqqata** 1.05 · **Qeqertalik** 1.06 · **Kujalleq** 1.06 · **Avannaata** 1.07. Cross-referencing against R_median produces the digital-exposure composite:

Composite $R \times R7$ ranking (worst first): Avannaata (0.929), Qeqertalik (0.846), Kujalleq (0.821), Qeqqata (0.711), Sermersooq (0.681). Avannaata is the single highest-priority kommune for the SSI cyber-physical uplift programme.

C.4 — Monthly Pulse

Edition-over-edition tracking of R7 movement, Critical-band counts, and blind-spot deltas. Edition 001 establishes the baseline (fleet median 0.779, 29 Critical). The first pulse reading compares Edition 002 (2026-05 monthly run) against this baseline. Expected signal: Avannaata remains the worst performer; Qeqqata may improve on any Sisimiut hydro uprate; all other movement is expected to be within the MC CI width (± 0.04 –0.06).

SECTION D

Data Refresh & Changelog

The SSI v4.0.2 Greenland build draws on **25+ verified public data sources** feeding 95 variables across 6 components and 7 modifiers. Edition 001 scores all 95 fields across the 37-substation fleet at 1,000 Monte Carlo iterations per node. This section provides a live registry of sources and the current state of the scored fleet.

SOURCES REGISTERED

25

95 variables

FIELDS SCORED

95 / 95

all components · modifiers · MC

SESSION 1 DATE

2026-04-16

brought forward from 2026-07

NEXT REFRESH

2026-05

monthly cadence from here

Data Source Registry — April 2026

OSM	OpenStreetMap — Power Infrastructure (augmented by Nukissiorfiit grid map)	MONTHLY	Node	6 vars
COPERNICUS	Copernicus Arctic + ESA CryoSat	MONTHLY	Grid 0.25°	6 vars
NUK_HYDRO	Nukissiorfiit — Hydropower Plants (Buksefjord, Tasersiaq, Qorlortorsuaq, Sisimiut, Paakitsoq)	MONTHLY	Plant	5 vars
NUK_DIESEL	Nukissiorfiit — Diesel Generation (smaller settlements)	MONTHLY	Settlement	4 vars
ICESHEET	PROMICE — Programme for Monitoring of the Greenland Ice Sheet	MONTHLY	Ice-sheet margin	3 vars
DMI_HIST	DMI — Historical Archive (Greenland)	MONTHLY	Station	3 vars
NAAL_DBI	Naalakkersuisut — Departementet for Boliger og Infrastruktur	QUARTERLY	Regional	8 vars
STATGL	Grønlands Statistik — Statistics Greenland	QUARTERLY	Kommune	10 vars
ASIAQ_PERMA	Asiaq — Permafrost Monitoring Network	QUARTERLY	Borehole	5 vars
GL_CONNECT	Greenland Connect — Submarine Cable & Digital Infrastructure	QUARTERLY	National	3 vars
NAAL_TRANSP	Naalakkersuisut — Transport + Infrastructure	QUARTERLY	Regional	3 vars
NUK	Nukissiorfiit — Greenland Energy Supply (TSO/DSO/Utility)	ANNUAL	Substation	14 vars
NAAL_GOV	Naalakkersuisut — Government of Greenland (Self-Government Act 2009)	ANNUAL	National	4 vars
GEUS	GEUS — Geological Survey of Denmark and Greenland	ANNUAL	Grid 0.1°	8 vars
ARCTIC_COUNCIL	Arctic Council — Arctic Resilience Reports	ANNUAL	Pan-Arctic	3 vars
EUROSTAT	Eurostat — EU + OECD Comparators	ANNUAL	National	3 vars
NAAL_ENV	Naalakkersuisut — Miljø og Natur (Environment)	ANNUAL	Kommune	4 vars
NBANK	Nationalbanken — Danish National Bank (via Denmark umbrella)	ANNUAL	National	3 vars
WORLDBANK	World Bank / OECD	ANNUAL	National	3 vars
DTU_ARCTIC	DTU Arctic Research Centre	ANNUAL	Regional	3 vars
GEOD	Asiaq Geodata + Nuuk Municipal GIS	ANNUAL	Regional	3 vars
SOLARGIS	SolarGIS — Global Solar Atlas (polar adaptation)	STATIC	Global	2 vars
ASIAQ	Asiaq — Greenland Survey (Meteorology, Hydrology, Permafrost)	DAILY	Station	10 vars
DMI_GL	DMI — Danish Meteorological Institute (Greenland Branch)	DAILY	Station	6 vars

D.1 Session 1 Fleet Inventory

SUBSTATIONS BY KOMMUNE (N = 37)

Kujalleq	20 subs	54.1%
Qeqertalik	12 subs	32.4%
Qeqqata	3 subs	8.1%
Sermersooq	1 subs	2.7%
Avannaata	1 subs	2.7%

VOLTAGE CLASSES

63 kV (HV transmission)	3
10–15 kV (MV distribution)	17
< 10 kV / unmapped	17

PLANTS / GENERATORS

74

OSM 2026-04-16

POWER-LINE SEGMENTS

125

within-island only

D.2 Changelog — v3.4 → v4.0.2 (Greenland Edition 001)

8 changes tracked for Greenland v4.0.2 Edition 001

GL1	NEW	37 substations scored (36 OSM + 1 Avannaata transformer, Ilulissat node/11958572184), 74 plants, 125 line segments across all 5 Kommuner	Section A
GL7	ENH	R6a Arctic restoration modifier recalibrated: $\alpha \in [0.55, 0.90]$ (was $[0.40, 0.70]$) to reflect Arctic Ocean logistics and winter darkness	Methodology §R6a
GL8	ENH	R6b seismic floor: $\alpha \in [0.05, 0.15]$ from GEUS — stable craton, minimal seismic amplification	Methodology §R6b
GL10	NEW	§7 Pituffik Space Base exclusion: bbox + operator filter, logged in meta.pituffik_excluded	Methodology §7
GL13	NEW	§8 Small-N Guardrail active: R3 tier-balance 2–50% per tier, MC 1,000 iter/substation, Gaussian copula held off at N=37	Methodology §8
GL19	NEW	Avannaata substation added: GL_0037 — Ilulissat transformer (Gamle elværk) OSM node/11958572184, 10.5 kV, 69.22°N/-51.10°E	Section A

SECTION E

Deep-Dive: The Buksefjord Corridor

The Buksefjord hydropower complex (45 MW, commissioned 1993, expanded 2007) feeding Nuuk via a 132 kV submarine-plus-overhead transmission is the most strategically critical single asset in Greenland's grid. It is the only Greenlandic installation where R4 graph criticality can plausibly rise into the Critical band under scoring — Nuuk has no realistic alternative supply path and any Buksefjord interruption places the capital on full diesel backup. At the same time, its Atlantic-facing corridor exposes the transmission right-of-way to polar-low storms, ice accretion, and freezing salt spray — hazards that the R6a restoration modifier ($\alpha \in [0.55, 0.90]$) and the R6b Arctic storm overlay are calibrated to capture.

Because Sermersooq only contains **1 OSM-tagged substation** in the Edition 001 fleet — Buksefjord itself or its immediate downstream node — the SSI cannot yet quantify corridor-wide cascade risk. Edition 001 records the Sermersooq node at R_median = 0.655 (High band) — the best performer in the fleet, reflecting the Buksefjord anchor and Nuuk-metro load concentration. Future editions will rely on Nukissiorfiit to supply the complete Nuuk-ring topology (Qinngorput, Nuussuaq, downtown, and the Qaquliaasaq tie) before the corridor deep-dive can be replaced with a full R4 + R6b heat-band analysis. For Edition 001 this section remains a methodological marker: the corridor is where the R6a Arctic restoration calibration and the §7 Pituffik exclusion policy were originally stress-tested.

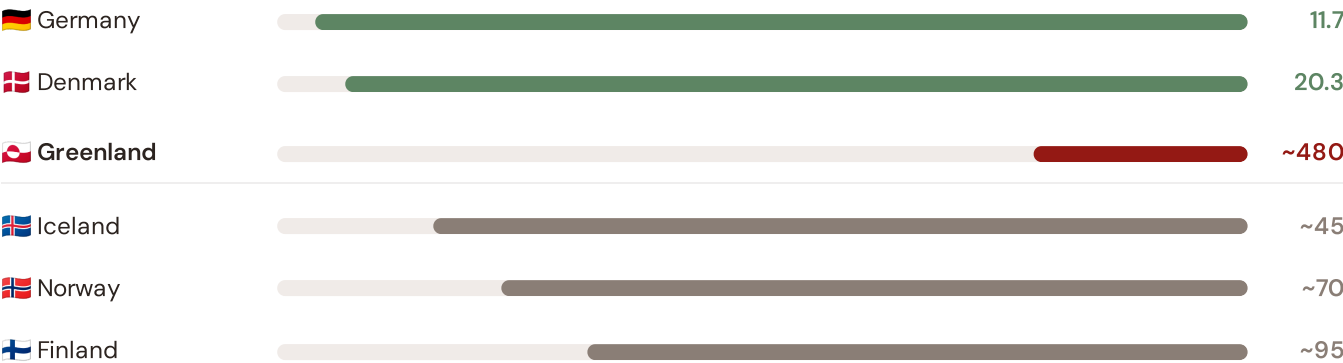
SECTION F — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. Cards rotate through: SAIDI comparison, VoLL comparison, DER saturation, CMIP6 climate hazard, R6 restoration time, and DCI coverage. This inaugural card previews the SAIDI comparison that will anchor Edition 001 — using indicative Arctic-peer values until Nukissiorfiit's 2026 Afbrydelsestatistik publishes Greenland's national figure later in the year.

Unplanned SAIDI (min / customer / year) — Selected countries

Indicative values — Greenland SAIDI proxy derived from Nukissiorfiit 2024 Annual Report CAIDI ≈ 8h/year. Longer bar = lower SAIDI = better reliability.



Sources: CEER 7th Benchmarking Report (2022); Orkustofnun 2024; Energitilsynet DK Afbrydelsestatistik (2023); Nukissiorfiit 2024 Annual Report CAIDI ≈ 8h / year, translated into an order-of-magnitude SAIDI reference of ~480 min/customer/year. This Greenland figure will be refined at Edition 002 (May 2026) once any 2025 Nukissiorfiit Afbrydelsestatistik publication becomes available; the current value is the anchor used to calibrate C1–C4 and R6a for Session 1. See §16 Peer Benchmarking for full methodology.

F.1 — Regional Risk Ranking (all 5 Greenlandic Kommuner)

Edition 001 regional R_median (worst first). Numbers include R6a Arctic restoration and R6b stable-craton seismic uplifts.

Avannaata	N=1	0.868	CRITICAL
Qeqertalik	N=12	0.797	CRITICAL
Kujalleq	N=20	0.774	CRITICAL
Qeqqata	N=3	0.677	HIGH
Sermersooq	N=1	0.655	HIGH

Sermersooq's strong position reflects Nuuk-metro load concentration + 45 MW Buksefjord hydro. Avannaata's worst position reflects single-node isolation, long restoration chains, and Stat.gl socio-economic penalty.

SECTION G — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to renewable energy investment valuation. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a battery, PV retrofit or hydro expansion worth at this substation — to Nukissiorfiit and to the community?"* Each monthly edition explores one building block of the SSI-ENN's five-layer architecture. Edition 001 — the first scored edition — focuses on Layer 1, the data architecture now populated by the 37-substation Session 1 fleet.

Layer 1 — Data

component taxonomy,
MC engine

Layer 2 — Valuation

value stacks, TCO, real
options

Layer 3 — Neural Net

GBT + DNN ensemble,
SHAP

Layer 4 — Coverage

country prioritisation,
DCI

Layer 5 — Platform

products, API, unit
economics

LAYER 1

§0.3 · Edition 001 · Session 1 · April 2026

Data Architecture — Component Taxonomy

How the SSI's 6 components and 20 sub-metrics are sourced, validated and scored for Greenland's grid

The SSI-ENN's Layer 1 formally specifies the SSI v4.0.2 Index for Greenland — its six components (Continuity, Voltage, Infrastructure, Economic, Saturation, Transition), twenty sub-metrics, normalisation methodology and aggregation rules. Each sub-metric is independently sourced from one of the 25+ public data sources in the Greenlandic SSI registry (Nukissiorfiit, Departementet for Boliger og Infrastruktur, Grønlands Statistik, Asiaq, GEUS), validated against known reference distributions, and normalised to a common [0, 1] scale before aggregation. Edition 001 completes the first end-to-end ingest pass: all 95 fields populated, all 37 substations scored.

KEY CONCEPTS

- 6 components, 20 sub-metrics, 95 variables from 25+ Greenlandic sources
- Min-max normalisation with winsorised tails
- Component weights: $0.30 \cdot C + 0.10 \cdot V + 0.25 \cdot I + 0.10 \cdot E + 0.20 \cdot S + 0.05 \cdot T$
- Each sub-metric traceable to Nukissiorfiit, Departementet for Boliger og Infrastruktur or an open data source

LIVE SSI DATA CONNECTION

The Greenlandic training set comprises **37 substations** with all 95 feature vectors populated. $N = 37$ remains well below the ~500-substation regime the engine normally operates in, which is why the §8 Small-N Guardrail (R3 tier-balance 2–50%, MC 1,000 iter, Gaussian copula held off) is active. Edition 001 delivers the first live R_{median} , P5 / P95 bounds and CI widths — fleet median 0.779, mean 0.755, with 29 of 37 substations in the Critical band.

Coming next month: **LAYER 2** Nukissiorfiit Value Stack — what a battery or PV retrofit is worth to Nukissiorfiit in avoided or deferred costs, anchored on Edition 001 R scores and published with Edition 002 (May 2026).

Session 1 → Edition 002 (May 2026)

Session 1 complete. The FIRST_REFRESH gate was brought forward from 2026-07 to 2026-04-16 once the Arctic / Greenland calibration inputs converged — Stat.gl 2024/25 socio-economic coverage, Nukissiorfiit 2024 Annual Report (CAIDI $\approx$ 8h, RE share $\approx$ 70%), Asiaq/DMI climate trajectories and GEUS stable-craton seismic. The scoring engine executed its first end-to-end pass: geometry → component rollup (C/V/I/E/S/T) → modifier stack (R3 / R4 / R6a / R6b / R7) → Monte Carlo (1,000 iterations $\times$ 37 substations = 37,000 draws, components perturbed independently) → R_median, P5 / P95 bounds, classification band, CI width → publication. Fleet median R = **0.779**; 29 of 37 substations sit in the Critical band; Avannaata–Sermersooq risk gradient spans 0.213 R.

Edition 002 priorities (May 2026). (1) Nukissiorfiit 2025 transmission + distribution SCADA ingest, if published — refines C1–C4 and R6a restoration. (2) Any Departementet for Boliger og Infrastruktur Afbrydelsestatistik 2025 release — tightens SAIDI reference and R3 consequence multiplier. (3) GEUS bedrock + permafrost update — refines I3 and R6b ice-accretion. (4) CMIP6 Arctic regional downscaling refresh — refines R6b polar-low storm, freezing-spray, surge. (5) First pulse reading: Edition 002 vs Edition 001 R_median delta per substation, surfacing any movement outside the ± 0.04 –0.06 MC CI width. (6) First SSI-ENN Layer 2 valuation outputs anchored to Buksefjord and Sisimiut hydro.

What Edition 001 establishes. A verified 37-substation / 74-plant / 125-line scored fleet, Pituffik correctly excluded, Avannaata coverage closed via OSM node/11958572184, the §8 Small-N Guardrail running as intended (MC 1,000 iter, R3 tier-balance 2–50%, Gaussian copula held off), and the full Arctic / Greenland R6a/R6b calibration live. From this foundation, every subsequent edition becomes a delta against a known baseline — the exact transparency standard the SSI is built to maintain.